

Digital Logics

BCA — Semester 1 — — Answer Sheet

Subject Code:

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group B

- 1. Design a combinational circuit with four inputs and one output. The output is equal to 1 when (i) all the inputs are equal to 1 or (ii) none of the inputs are equal to 1 or (iii) an odd number of inputs are equal to 1.**

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

- 2. Implement the following function: $F = \sum(0,1,3,4,5,8,9,10,15)$ using i) Decoder ii) Multiplexer iii) PLA**

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

- 3. What do you mean by race condition in JK Flip Flop and mention the methods to overcome it? Explain Master Slave Flip-flop using JK flip flop with logic circuit, truth table and timing diagram.**

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

- 4. Convert (1011.110) into decimal and hexadecimal.**

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

- 5. Convert (51966.57) into octal and hexadecimal number system.**

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

- 6. Subtract (111000.110) — (110100.101) using both 2's and 1's complement.**

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

- 7. State and prove De-Morgan's Theorems.**

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Define Half-subtractor with truth table and logic diagram.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. What is decoder? Implement 8 x 1 MUX using 4 x 1 MUX.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. What is clocked RS flip-flop? Explain with logic diagram and characteristic table.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Design Mod-3 synchronous counter.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Draw a Parallel-In Parallel-Out Shift register and explain it.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.